

Zach Sweeney → Exam 1

Problem 1

Given $G = 0.19$ $P = 14.3 \text{ g/cc}$ $\sigma_p = 587 \cdot 10^{-24} \text{ cm}^2$
 $\phi = 4 \cdot 10^{12}$
 Find Q
 U-N fuel

$$Q = E_f \cdot \underbrace{N_p \cdot \sigma_p \cdot \phi}_{N_{235}}$$

$$N_{235} = \frac{P N_A G}{M_{UN}}$$

$$M_{UN} = (0.19 \cdot 235 + 0.81 \cdot 238) + 14$$

$$M_{UN} = 251.43$$

$$N_{235} = \frac{14.3 \text{ g/cc} \cdot 6.022 \cdot 10^{23} \frac{\text{atoms}}{\text{mol}} \cdot \frac{1 \text{ atom U}}{1 \text{ molecule UN}} \cdot 0.19}{251.43}$$

$$N_{235} = 6.51 \cdot 10^{21} \frac{\text{U235}}{\text{cm}^3}$$

$$\therefore Q = E_f \cdot N_p \cdot \sigma_p \cdot \phi \rightarrow \text{200}$$

$$Q = 200 \cdot 10^6 \text{ eV} \cdot 1.6 \cdot 10^{-19} \frac{\text{kJ}}{\text{eV}} \cdot 6.51 \cdot 10^{21} \cdot 587 \cdot 10^{-24} \cdot (4 \cdot 10^{12} \frac{1}{\text{cm}^2 \cdot \text{s}})$$

$$Q = 4189 \frac{\text{W}}{\text{cc}}$$

Part 2 — For UO_2 , $P = 10.97$

$$N_{235} = \frac{P N_A G}{M} = \frac{10.97 \cdot 6.022 \cdot 10^{23} \cdot X}{M_{\text{UO}_2}}$$

$$M_{\text{UO}_2} = (X \cdot 235) + (1-X) \cdot 238 + 2 \cdot 16$$

$$N_{235} = \frac{10.97 \cdot 6.022 \cdot 10^{23} \cdot X}{235X + 238 - 238X + 32}$$

Problem 1 (cat)

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Simplifying $N_{235} \rightarrow N_{235} =$

$$\frac{6.61 \cdot 10^{24} \cdot X}{-3X + 270}$$

where $X = \text{enrichment}$

Now, plugging into heat eqn:

$$Q = E_F N_F \Sigma_F \phi \rightarrow N_F = \frac{Q}{E_F \phi \Sigma_F}$$

$$\frac{6.61 \cdot 10^{24} X}{-3X} + \frac{X}{270} = \frac{Q}{E_F \phi \Sigma_F}$$

$$X = 270 \cdot \left(\frac{Q}{E_F \phi \Sigma_F 2.203 \cdot 10^{21}} \right)$$

Plugging in $\rightarrow$

$$X = 270 \cdot \frac{489 \frac{\text{W}}{\text{cc}}}{200 \cdot 10^6 \cdot 1.6 \cdot 10^{-14} \cdot 587.10^{21} \cdot 4 \cdot 10^{12}}$$

$X = 0.176$

 $\rightarrow$ this seems too high

Problem 2

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Given $k_c = 0.18$ $k_p = 0.01$ $h_{cool} = 1.5$

$Q = 350$ $R_f = 0.45$ $t_g = 80 \mu m$ $t_c = 0.06 cm$

$T_{cool} = 540 K$

T_{cool} is known, work outside-in

$$T_{oc} - T_{cool} = \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}}$$

First, calculate LHR $\rightarrow LHR = \pi R^2 Q$

$$LHR = \pi (0.45)^2 350$$

$$LHR = 222 \frac{W}{cm^2}$$

$$T_{oc} = T_{cool} + \frac{LHR}{2\pi R_f h_{cool}} = \frac{222}{2\pi \cdot 0.45 \cdot 1.5} + 540 K$$

$$T_{oc} = 592 K$$

$T_{oc} \rightarrow T_{midpt} \rightarrow$

here $t_c = t_g - r$

$$T_{ic} - T_{oc} = \frac{\cancel{LHR}}{\cancel{2\pi R_f h_{cool}}} \frac{LHR}{2\pi R_f} \frac{t_g}{k_c}$$

$$T_{o.1} = T_{oc} + \frac{LHR}{2\pi R_f} \cdot \frac{t_{o.1}}{k_c}$$

midpoint of cladding

$$T_{o.1} = 592 K + \frac{222 \frac{W}{cm^2}}{2\pi \cdot 0.45} \cdot \frac{0.03}{0.18}$$

$$T = 697 K$$

Problem 3 $L = 3.6 \text{ m}$

Given $LHR^0 = 400 \frac{\text{W}}{\text{cm}}$ $\gamma = 1.25$

$C_p = 4200 \frac{\text{J}}{\text{kgK}}$ $\dot{m} = 0.1 \frac{\text{kg}}{\text{s}}$ $T_i = 580 \text{ K}$

Find LHR @ $z = 2.1 \text{ m}$ (Axial ~~heat~~ conv)

$$z_0 = \frac{L}{2} = \frac{3.6}{2} = 1.8 \text{ m}$$

$$LHR \frac{z}{z_0} = LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{z_0} - 1 \right) \right]$$

$$LHR(2.1) = 400 \cdot \cos \left[\frac{\pi}{2(1.25)} \left(\frac{2.1}{1.8} - 1 \right) \right]$$

$$LHR(2.1) = 391 \frac{\text{W}}{\text{cm}}$$

Now Find T_{cool}

$$T_{\text{cool}} - T_{\text{cool}}^i = \frac{2\gamma}{\pi} \cdot \frac{z_0 LHR^0}{\dot{m} C_p} \left\{ \sin \frac{\pi}{2\gamma} + \sin \left(\frac{\pi}{2\gamma} \left(\frac{z}{z_0} - 1 \right) \right) \right\}$$

at outlet, $z = 3.6$

$$T_{\text{cool}}^{\text{out}} = T_{\text{cool}}^i + \frac{2\gamma}{\pi} \cdot \frac{z_0 LHR^0}{\dot{m} C_p} \left\{ \sin \left(\frac{\pi}{2\gamma} \right) + \sin \left(\frac{\pi}{2\gamma} \left(\frac{z}{z_0} - 1 \right) \right) \right\}$$

Plugging in $\rightarrow$

$$T_{\text{cool}}^{\text{out}} = 580 \text{ K} + \frac{2 \cdot 1.25}{\pi} \cdot \frac{1.8 \cdot 400}{0.1 \cdot 4200} \left\{ \sin(1.25) + \sin \left(\frac{\pi}{2 \cdot 1.25} \left(\frac{3.6}{1.8} - 1 \right) \right) \right\}$$

$$T_{\text{out}} = 580 \text{ K} + 25.9 \text{ K} = 606 \text{ K} \rightarrow \text{coolant outlet}$$

Seems like a very small ΔT , could be units error

Problem 4Given, 5% bump FIMA, $T_f = 1200\text{ K}$

w/ FIMA

$$K_{\text{ox}} = \frac{1}{A + BT}$$

$$A = 3.8 + 200 \cdot 0.05 = 13.8$$

$$B = 0.0217$$

$$K_{\text{ox}} = \frac{1}{13.8 + 0.0217 \cdot 1200\text{ K}}$$

$$K_{\text{ox}} \text{ w/ FIMA} = 0.0251 \frac{\text{W}}{\text{cm}^2\text{K}}$$

w/o FIMA $\rightarrow$ ~~K_{ox}~~

$$A = 3.8 + 200 \cdot \overset{\rightarrow 0}{\text{FIMA}} = 3.8$$

$$K_{\text{ox}} = \frac{1}{3.8 + 0.0217 \cdot 1200}$$

$$K_{\text{ox}} = 0.0335 \frac{\text{W}}{\text{cm}^2\text{K}} \quad (\text{No FIMA})$$

$\therefore$ FIMA accounts for about a ~~10%~~ reduction
 $\hookrightarrow 25-35\%$

5) 6.) Steady-state, Axisymmetric, Constant T_i ,
and k is independent of T

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6.) UF_6 is used for enrichment. For centrifuge, gas is flown into rotating vessel, where enriched U-235 flows to the center and is extracted. Depleted U is heavier, moves to outside.

7.) DNBR is Ratio of critical heat flux to Actual heat flux.
Critical heat flux is when flow is in transition boiling. ~~###~~ This

8.) σ_{max} density is a ratio of fuel volume to cladding internal volume. This ensures proper heat flow, and that cladding or gap are sufficient size to allow proper flow.

a.) Swelling, Creep

10.) Cladding $\rightarrow$ contains fuel pellets in uniform space
 $\hookrightarrow$ contains fission products. ^{somewhat...}

Zr is effective since it has a low neutron capture cross-section.

11.) Fuel Rod, cladding, gap, ~~some~~ coolant

$\hookrightarrow$ technically, more of a secondary role

- 12.) 1.) how effective fuel transfers heat to coolant
2.) behavior during Normal operations
3.) behavior during accidents / Accident tolerance

13.) Finite Element $\rightarrow$ Any geometry / BC
 $\rightarrow$ Complicated

Finite Volume $\rightarrow$ Can model any geometry
 $\rightarrow$ Limited on BC's

Finite Difference $\rightarrow$ simplest
 $\rightarrow$ can only handle basic geometry

14.) $\oplus$ Very Inexpensive
Years of OE, predictable

$\ominus$ Brittle
Not the best thermal conductor

15.) ~~Implicit~~ $\rightarrow$ solves using present data point
~~Explicit~~ $\rightarrow$ solves
 $\rightarrow$ backward Euler
Implicit $\rightarrow$ future data point used
 $\rightarrow$ Forward Euler
explicit $\rightarrow$ current state used to start